A Task-Level Provider Utility Model

Matthew Pollard

Provider selection should minimize the expected risk-adjusted cost of completing an entire task, not the nominal price of an isolated request. A task consists of N conversational turns whose reusable prefix generally grows over time. The model includes cached and uncached input charges, output-token charges, latency and token-generation time converted into dollars, endpoint failures and retries, stochastic cache misses, shrinkage of noisy endpoint statistics, and a variance penalty for operational and estimation risk.

1 Conversation structure

For turn $k = 1, \ldots, N$, let

- A_k be newly appended prompt tokens, such as a user message or tool result;
- O_k be completion tokens generated on the turn; and
- S_k be reusable prefix tokens preceding the new input.

For a conversation that resubmits its complete history,

$$S_k = \sum_{j<k} (A_j + O_j). \tag{1}$$

The complete prompt length is therefore

$$P_k = S_k + A_k. \tag{2}$$

This distinction matters: A_k is necessarily new input, whereas only S_k is eligible for a cache hit.

2 Endpoint parameters

For endpoint e, let

- w_e be the uncached-input or cache-write price per token;
- r_e be the cached-read price per token;
- c_e be the completion price per token;
- f_e be the fixed request charge;
- h_e be the probability that the reusable prefix is cached;
- $p_e = 1 - h_e$ be the cache-miss probability;
- u_e be the probability that an individual attempt succeeds;
- L_e be time to first token in seconds;
- G_e be token-generation throughput in tokens per second;
- T_e^F be time consumed by a failed attempt;
- $\beta_e \in [0, 1]$ be the fraction of attempted monetary input cost charged on failure; and
- v be the user's value of time in USD per hour.

Define the dollar value of one second as

$$\rho = \frac{v}{3600}. \tag{3}$$

The successful-turn wall-clock time is

$$T^S_{e,k} = L_e + \frac{O_k}{G_e}. \tag{4}$$

The first term values latency to first token; the second values token-generation time.

3 Cache-miss model

For every attempt j at turn k, define

$$M_{k,j} \sim \text{Bernoulli}(p_e), \tag{5}$$

where $M_{k,j} = 1$ denotes a complete miss of the reusable prefix. Cache events are assumed independent across attempts and turns and independent of endpoint availability.

Let

$$\Delta_e = w_e - r_e \tag{6}$$

be the cache-miss price spread. The attempted input cost is

$$I_{e,k,j} = f_e + w_e A_k + S_k \left[r_e + M_{k,j} \Delta_e \right]. \tag{7}$$

Thus new tokens A_k are always charged at w_e. On a hit, the reusable prefix S_k is charged at r_e; on a miss, the same prefix is charged at w_e.

For a fixed hit probability,

$$\begin{aligned} \bar{I}_{e,k} &= \mathbb{E}[I_{e,k,j}] && (8) \\ &= f_e + w_e A_k + S_k \left[r_e + p_e \Delta_e \right], && (9) \end{aligned}$$

and

$$V^I_{e,k} = \text{Var}(I_{e,k,j}) = S_k^2 \Delta_e^2 p_e (1 - p_e). \tag{10}$$

4 Uptime and retry model

Let an attempt succeed independently with probability u_e. The number of failed attempts before the first success is

$$K_{e,k} \sim \text{GeometricFailures}(u_e), \tag{11}$$

with

$$\mathbb{E}[K_{e,k}] = \frac{1 - u_e}{u_e}, \qquad \text{Var}(K_{e,k}) = \frac{1 - u_e}{u_e^2}. \tag{12}$$

This models retrying the endpoint until one attempt succeeds. An implementation may replace this with an explicit fallback chain when the fallback endpoint is known.

The cost of the successful attempt is

$$Y^S_{e,k} = I_{e,k} + c_e O_k + \rho T^S_{e,k}. \tag{13}$$

The cost of a failed attempt is

$$Y_{e,k}^F = \beta_e I_{e,k} + \rho T_e^F. \tag{14}$$

The parameter β_e makes the billing assumption explicit. Setting $\beta_e = 0$ means failed attempts incur time cost but no token charge, whereas $\beta_e = 1$ conservatively assigns them their full attempted input charge.

The completed-turn cost is

$$X_{e,k} = Y_{e,k}^S + \sum_{j=1}^{K_{e,k}} Y_{e,k,j}^F. \tag{15}$$

Define

$$B_e = \frac{1 - u_e}{u_e}. \tag{16}$$

Then

$$\mu_{e,k}^S = \bar{I}_{e,k} + c_e O_k + \rho \left(L_e + \frac{O_k}{G_e} \right), \tag{17}$$

$$\mu_{e,k}^F = \beta_e \bar{I}_{e,k} + \rho T_e^F, \tag{18}$$

and the expected cost of completing turn k is

$$\boxed{\mathbb{E}[X_{e,k}] = \mu_{e,k}^S + B_e \mu_{e,k}^F}. \tag{19}$$

Equivalently,

$$\boxed{\begin{aligned} \mathbb{E}[X_{e,k}] = \bar{I}_{e,k} + c_e O_k + \rho \left(L_e + \frac{O_k}{G_e} \right) \\ + \frac{1 - u_e}{u_e} \left(\beta_e \bar{I}_{e,k} + \rho T_e^F \right). \end{aligned}} \tag{20}$$

Unlike a simple $(1 - u_e)$ surcharge, the factor $(1 - u_e)/u_e$ accounts for the possibility of more than one failed attempt.

5 Task-level expected cost

The task cost is the sum of its completed-turn costs:

$$X_e^{\text{task}} = \sum_{k=1}^{N} X_{e,k}. \tag{21}$$

Therefore,

$$\boxed{\mu_e^{\text{task}} = \mathbb{E}[X_e^{\text{task}}] = \sum_{k=1}^{N} \left[\mu_{e,k}^S + B_e \mu_{e,k}^F \right]}. \tag{22}$$

This expected dollar cost per completed task should be the primary ranking quantity.

6 Constant-growth conversation

For intuition, suppose every turn adds a new prompt tokens and produces o completion tokens. Let

$$d = a + o, \qquad S_k = (k-1)d. \tag{23}$$

Define

$$\pi_e = r_e + p_e(w_e - r_e) = h_e r_e + (1 - h_e) w_e, \tag{24}$$

$$b_e = f_e + w_e a, \tag{25}$$

$$y_e = c_e o + \rho \left(L_e + \frac{o}{G_e} \right), \tag{26}$$

$$z_e = \rho T_e^F, \tag{27}$$

and

$$D_e = 1 + \beta_e \frac{1 - u_e}{u_e}. \tag{28}$$

Then the expected task cost has the closed form

$$\boxed{\mu_e^{\text{task}} = N \left[D_e b_e + y_e + B_e z_e \right] + D_e \pi_e d \frac{N(N-1)}{2}}. \tag{29}$$

The first term is linear in the number of turns. The second is quadratic because the prefix grows while being submitted repeatedly.

The expected cache-miss premium, excluding the ordinary cached-read charge, is

$$\boxed{D_e p_e (w_e - r_e) d \frac{N(N-1)}{2}}. \tag{30}$$

Consequently, a 5% cache-miss rate is not merely a 5% task-level surcharge. It applies to a sequence of progressively larger prefixes.

7 Operational variance

Assume successful and failed attempt times have variances

$$\text{Var}(T_{e,k}^S) = (\sigma_{e,k}^S)^2, \qquad \text{Var}(T_e^F) = (\sigma_e^F)^2. \tag{31}$$

Then

$$V_{e,k}^S = V_{e,k}^I + \rho^2 (\sigma_{e,k}^S)^2, \tag{32}$$

and

$$V_{e,k}^F = \beta_e^2 V_{e,k}^I + \rho^2 (\sigma_e^F)^2. \tag{33}$$

Using the variance of a random sum,

$$\boxed{\text{Var}(X_{e,k}) = V_{e,k}^S + B_e V_{e,k}^F + \frac{1 - u_e}{u_e^2} \left(\mu_{e,k}^F \right)^2}. \tag{34}$$

The three components are uncertainty in the successful attempt, uncertainty in the costs of failed attempts, and uncertainty in how many failures occur before success.

Under independence across turns,

$$\boxed{V_{e,\text{process}}^{\text{task}} = \sum_{k=1}^{N} \text{Var}(X_{e,k})}. \tag{35}$$

For constant transcript growth,

$$\sum_{k=1}^{N} S_k^2 = d^2 \frac{N(N-1)(2N-1)}{6}. \tag{36}$$

Accordingly, cache-miss variance grows as $O(N^3)$, and its standard deviation grows as $O(N^{3/2})$.

8 Shrinkage of cache hit rate and uptime

Raw 24-hour endpoint statistics can be unstable for low-volume endpoints. Both cache hit rate and uptime should therefore be shrunk toward population priors.

For cache hits, let $\hat{h}_e$ be the published cache-hit rate, n_e^h the effective cache sample size, h_0 the population prior mean, and κ_h the prior strength. Use the beta prior

$$h_e \sim \text{Beta}\left(\kappa_h h_0,\, \kappa_h(1-h_0)\right). \tag{37}$$

After observing effective evidence n_e^h, the posterior parameters are

$$\alpha_e^h = \kappa_h h_0 + n_e^h \hat{h}_e, \tag{38}$$

and

$$\beta_e^h = \kappa_h(1-h_0) + n_e^h(1-\hat{h}_e). \tag{39}$$

The shrunk hit rate is

$$\boxed{\tilde{h}_e = \frac{\kappa_h h_0 + n_e^h \hat{h}_e}{\kappa_h + n_e^h}}. \tag{40}$$

Its posterior variance is

$$\boxed{s_{h,e}^2 = \frac{\alpha_e^h \beta_e^h}{(\alpha_e^h + \beta_e^h)^2(\alpha_e^h + \beta_e^h + 1)}}. \tag{41}$$

The effective sample size should not simply equal raw token volume. Cache outcomes within one request or conversation are correlated. A practical approximation is

$$n_e^h = \frac{T_e}{B_h}, \tag{42}$$

where T_e is relevant prompt-token volume and B_h is a configurable correlation-block size.

Uptime is treated analogously. Given published uptime $\hat{u}_e$, effective sample size n_e^u, prior mean u_0, and prior strength κ_u,

$$\boxed{\tilde{u}_e = \frac{\kappa_u u_0 + n_e^u \hat{u}_e}{\kappa_u + n_e^u}}. \tag{43}$$

Its posterior variance $s_{u,e}^2$ follows the same beta-posterior expression. Uptime evidence should be measured in requests or approximately independent time buckets, not tokens.

The expected-cost and process-variance equations use the shrunk values

$$h_e = \tilde{h}_e, \qquad u_e = \tilde{u}_e. \tag{44}$$

9 Shrinkage of latency and generation speed

Latency and generation speed can also be unstable at low volume. Because both quantities are positive and usually right-skewed, shrink them in log space. For a positive performance parameter θ_e, such as TTFT or seconds per generated token, let $\hat{\theta}_e$ be the endpoint estimate and θ_0 the model-wide or provider-wide prior. Define

$$\log \tilde{\theta}_e = \omega_e \log \hat{\theta}_e + (1 - \omega_e) \log \theta_0, \tag{45}$$

where

$$\omega_e = \frac{n_e^\theta}{n_e^\theta + \kappa_\theta}. \tag{46}$$

For throughput, shrink seconds per token $1/G_e$, rather than throughput G_e, because time cost is linear in $1/G_e$.

When only p50 values are available, the resulting time cost is a point estimate. When multiple quantiles or raw observations are available, they should also be used to estimate σ^S and σ^F.

10 Parameter-estimation risk

Process variance assumes that the endpoint parameters are known. Shrinkage reduces estimation error but does not eliminate it. Let

$$\boldsymbol{\theta}_e = (h_e, u_e, L_e, G_e, T_e^F, \ldots) \tag{47}$$

have posterior covariance matrix $\Sigma_{\theta,e}$. A first-order approximation to uncertainty in expected task cost is

$$\boxed{V_{e,\text{parameter}}^{\text{task}} \approx \nabla_\theta \mu_e^{\text{task}\,\mathsf{T}} \Sigma_{\theta,e} \nabla_\theta \mu_e^{\text{task}}}. \tag{48}$$

For independent parameter estimates, this reduces to a sum of squared sensitivities multiplied by their variances.

Useful sensitivities include

$$\frac{\partial \mu_e^{\text{task}}}{\partial h_e} = -\left(1 + \beta_e \frac{1 - u_e}{u_e}\right)(w_e - r_e) \sum_{k=1}^{N} S_k, \tag{49}$$

$$\frac{\partial \mu_e^{\text{task}}}{\partial u_e} = -\frac{1}{u_e^2} \sum_{k=1}^{N} \mu_{e,k}^F, \tag{50}$$

$$\frac{\partial \mu_e^{\text{task}}}{\partial L_e} = \rho N, \tag{51}$$

and, when throughput is treated as a common endpoint parameter,

$$\frac{\partial \mu_e^{\text{task}}}{\partial G_e} = -\rho \frac{\sum_{k=1}^{N} O_k}{G_e^2}. \tag{52}$$

For constant transcript growth,

$$\sum_{k=1}^{N} S_k = O(N^2). \tag{53}$$

Therefore uncertainty in the endpoint's true cache-hit rate contributes

$$V_{h,e}^{\text{parameter}} = O(N^4), \tag{54}$$

even though independent realized cache misses contribute only $O(N^3)$ process variance. Equivalently, realized miss-risk standard deviation grows as $O(N^{3/2})$, while uncertainty about the true miss rate can produce an $O(N^2)$ standard-deviation penalty. This is why uncertain cache statistics become especially consequential for long agent trajectories.

11 Risk-adjusted provider score

The final provider score is

$$\boxed{J_e = \mu_e^{\text{task}} + \lambda_{\text{process}}\sqrt{V_{e,\text{process}}^{\text{task}}} + \lambda_{\text{parameter}}\sqrt{V_{e,\text{parameter}}^{\text{task}}}}. \tag{55}$$

Here λ_{process} penalizes unpredictable realized costs and delays, while $\lambda_{\text{parameter}}$ penalizes poorly estimated endpoint statistics. Reasonable interpretations are

- $\lambda_{\text{process}} = \lambda_{\text{parameter}} = 0$: risk-neutral expected cost;
- $\lambda = 1$: a one-standard-deviation penalty; and
- $\lambda \approx 1.645$: approximate one-sided 95% protection under a normal approximation.

Separating the penalties is preferable because a high-volume endpoint can have meaningful operational variance but little estimation uncertainty, while a low-volume endpoint may look excellent solely because its statistics are poorly estimated.

12 Reported outputs

For each endpoint, the tool should report

1. expected dollar cost per completed task;
2. risk-adjusted dollar score J_e;
3. expected number of failed attempts;
4. expected uncached-input, cached-read, and output tokens;
5. dollar cost of TTFT and token generation;
6. cache-miss premium;
7. uptime-retry premium;
8. process standard deviation;
9. parameter-estimation standard deviation; and
10. raw and shrunk cache-hit and uptime estimates.

The primary ranking should use dollars per completed task. A normalized per-million-token figure may be shown secondarily, but it must state whether the denominator counts unique task tokens or repeatedly submitted prompt tokens.